Question	Answer
what is closed weigths vs open?	The closed weights are fixed. The pre-trained model parameters cannot be inspected or modified by users. The open weights are publicly available model parameters that can be downloaded, fine-tuned, and run independently.
How can we measure Benchmark scores of individual LLMs ?	Measure an LLM by evaluating it on standardized benchmark datasets and comparing metrics such as accuracy, latency, throughput, and cost.
What is the difference between Foundational Model and Large Language Models?	A foundation model is a broad, general-purpose pretrained model that can support multiple tasks, while an LLM is a foundation model specifically designed to understand and generate human language.
What do you mean by weights of models?	Model weights are like the model's learned "memory" or adjustable settings that help it recognize patterns and produce answers.
In this exective course, will we use Open Weight models or Frontier models?	Frontier models
are specialized ones build on top of frontier ones with fine tuning?	Often yes. The specialized models are created by fine-tuning or otherwise adapting a frontier foundation model for a specific domain or task, though some are trained from scratch.
opus, hiku febble detials?	Claude Opus: Anthropic's most capable model family, suited to complex reasoning, coding, and difficult analysis. Claude Haiku: A smaller, faster, and generally cheaper model, suited to high-volume, low-latency tasks.
the 'frontier" model list is dated. with the speed at which evolutions are coming out weekly, will we be keeping pace in this course?	rontier models evolve rapidly, any model list can become outdated quickly; it is better to treat examples as time-specific and check current provider announcements or benchmarks for the latest models.
Sir, what are these percentages in the capability benchmarks table?	percentage of benchmark questions each model answered correctly—for example, GPT-4.1's 90% on HumanEval means it successfully completed about 90% of the 164 coding problems.
Hi. Why are we not comparing the latest models?	We should compare the latest models for current relevance, but benchmark results may lag because models, versions, access, evaluation setups, and reliable test results are not updated simultaneously.
MMLU, HumanEval - these are again models used for Eval?	MMLU and HumanEval are benchmark datasets and evaluation suites, not models, used to measure the performance of AI models on knowledge and coding tasks, respectively.
Why do LLM providers have different pricing for input and output tokens? What is it that output tokens are expensive?	Output tokens cost more because generating them requires step-by-step computation for each token, while input tokens can often be processed more efficiently in parallel and may benefit from caching.
Capablity: How can we do benchmarking? or Where can we find this benchmarks?	You can benchmark an LLM by running it on standardized datasets from Hugging Face, LMSYS Chatbot Arena, HELM, Papers with Code, and provider documentation, then comparing accuracy, latency, cost, and safety.
Isn't MMLU becoming saturated as a benchmark? because every model now scores above 90%? GPQA Diamond, SWE-bench Verified, and ARC-AGI-2, What about these benchmarks?	Yes—MMLU is becoming saturated, so newer benchmarks such as GPQA Diamond (expert-level reasoning), SWE-bench Verified (real-world software engineering), and ARC-AGI-2 (novel abstract reasoning) provide more challenging and informative evaluations.
We have agents configured in pipeline, how can we evaluate its performance !	Evaluate the agent using task success rate, factual accuracy, tool-call correctness, safety, latency, cost, and human review on representative real-world test cases.
for automotive SW code generation, requirement analysis, validation what LLM model is suitable?	For automotive software, use a strong coding/reasoning model—such as Claude Opus/Sonnet, GPT-5, or Gemini Pro—with RAG and human verification, while validating against ISO 26262, ASPICE, MISRA, and cybersecurity requirements.
can we create our own benchmark test for domain specific knowlegde? and suppose we use a fine tunned model for an app related to healthcare, but we are testing the model on ai generated data, will this model results be trustable? or will there be baised as a lot of models are trained on same data?	Yes, you can create a domain-specific benchmark, but AI-generated test data alone is not sufficient for trustworthy healthcare evaluation because it may contain errors, bias, and overlap with the model's training data.
Is there a difference between server hosted and cloud hosted? WHat does hostinger provide among the two?	Server-hosted usually means running on a dedicated or on-premises server, while cloud-hosted runs on provider-managed, scalable infrastructure; Hostinger mainly provides cloud hosting and traditional shared/VPS hosting, depending on the plan.
Does context include the prompt we are giving? Like will larger prompts lead to token exhaustion? or context is only code base.	Yes, context includes your prompt, conversation history, uploaded files, and retrieved code, so larger inputs consume more tokens and can reduce the space available for the model's response.
We have agents configured in pipeline, how can we evaluate its performance ! if we are using the Human Eval , do we evaluate it against the documents, won't it bloat the context so , how do we mitigate it up !	For pipeline agents, evaluate task success, factuality, tool-call accuracy, safety, latency, cost, and consistency using a representative test set. HumanEval is for standalone code generation—not document-grounded agents; evaluate against expected outputs and requirements using retrieval, chunking, summaries, metadata filters, and automated/unit tests, rather than placing the entire document in the prompt.

is there any tools available to test the Hallucination ? or how to check the best model interms of Hallucination?	Yes—tools such as Ragas, DeepEval, TruLens, Arize Phoenix, and LangSmith can evaluate hallucinations by checking answers against trusted reference documents, citations, and expected outputs.
what are use cases for llama	Llama models are used for applications such as chatbots, coding assistants, document summarization, question answering, content generation, and domain-specific AI systems through fine-tuning.
for automotive SW code generation, requirement analysis, validation what open weight LLM model is suitable?	For automotive software, consider Qwen3-Coder, DeepSeek-Coder, or Llama 4, using LoRA/QLoRA, RAG, and strict human review against ASPICE, ISO 26262, MISRA, and AUTOSAR requirements.
How we can fine tune LLM?	You fine-tune an LLM by training it further on carefully prepared examples from your domain, using methods such as LoRA/QLoRA, then evaluating it on separate unseen data before deployment.
Does training the model means adding documents of information to its context? how is itdifferent from RAG approach	Training changes the model's internal weights using examples, while RAG retrieves relevant documents at query time and adds them to the prompt without changing the model.
Why are we referring to gpt-4.1 specifically? Are the latest ones (eg:5.3) not effective?	GPT-4.1 may be referenced because the material used it as a fixed benchmark example; newer models may perform better, but comparisons are valid only when the model version, prompt, tools, and evaluation setup are controlled.
can we train a model locally on our system and can run internally in our system locally and perform different task and once we download the model , does it have a cutoff training date and is it possible to upgrade to the other model from same vendor and if we do this do we have an option to copy or export trained input data somehow ??	Yes, you can run an open-weight model locally, but training requires substantial GPU resources; downloaded models have a fixed training-data cutoff, upgrades require downloading and deploying a newer model, and your fine-tuning data can usually be retained/exported separately because it is not stored directly in the base model.
What are the full form of LoRA and GGUF?	LoRA means Low-Rank Adaptation, while GGUF is a model file format used to store and run quantized LLMs efficiently, especially with tools such as llama.cpp.
Is Open source models more favorable if we do not want to expose our source code	Yes, self-hosted open-weight models can be preferable because your source code can remain inside your infrastructure, but you must still secure logs, prompts, tools, monitoring, backups, and access controls.
If I run a model in my local machine, example a macbook pro, will it have any long term side-effects because I am running them continously?? will it degrade over time??	Running an LLM continuously on a MacBook Pro usually won't degrade the model, but prolonged heat and battery charging can accelerate hardware or battery wear, so use cooling, ventilation, power-management settings, and avoid sustained overheating.
Even if we use Open weight self hosted, we have to bill our clients. so the token should be charged rt? How do we decide on how much we charge to the client?	Yes, you can charge clients based on input/output tokens, requests, compute time, or a subscription, using your infrastructure cost plus operations, support, margin, and a contingency; track usage and publish clear pricing and limits.
Then how do we regularly train Open weights?	You usually do not retrain the base model regularly; instead, update knowledge with RAG, and periodically fine-tune adapters such as LoRA using newly reviewed, high-quality examples.
how open weight models can be trained or fine tuned time to time? if downloaded on local machine	A downloaded open-weight model can be periodically fine-tuned locally by preparing new, high-quality examples, training a LoRA/QLoRA adapter, evaluating it on unseen tests, and saving or deploying the updated adapter alongside the original model.
Please also try to give use case and live example of LoRA and GGUF?	LoRA use case Use case: Adapt Llama to generate automotive software requirements in your company's preferred format without retraining the entire model. Example: You fine-tune a Llama model with 5,000 approved requirement examples: Input: "Create a requirement for automatic emergency braking." Output: "The system shall detect…" in your required template. LoRA stores only a small adapter file containing the changes, while the original model remains unchanged. GGUF use case Use case: Run a quantized Llama model locally on a MacBook without a large GPU.
i mean, doesn't open weight models require huge amount of memory and gpus for fine tunning and for local use! Also for data sensitivity for clinets/users, cant we hash/encrpyt their names and send them to api so that their personal info dont get exposed and when the llm generates a result, we can match key or just dencrpy of soemthing that data?	Yes—fine-tuning generally needs more memory/GPU than inference, but smaller quantized models and LoRA/QLoRA can make local use practical. For sensitive data, hashing or encryption can help, but plain hashing is not sufficient for names because names are easy to guess; use tokenization/pseudonymization with random, securely stored mappings, send only the minimum necessary data, and never send encryption keys to the LLM API.

what do we mean by token	A token is a small piece of text—such as a word, part of a word, punctuation mark, or space—that an LLM converts into numbers so it can process and generate language.
are there open source solutions that can nearly estimate the num of tokens that gets consumed before sending Prompt API call to the model.. Atleast 90% accuracy	Yes—use the model's official tokenizer locally; tools such as TikToken, Hugging Face Tokenizers, and SentencePiece can usually estimate token usage very accurately, often close to 100%, before making the API call.
sir explain how these words are distributed to token here	Yes—when you send text, the tokenizer splits it into tokens, which may be whole words, parts of words, punctuation, or spaces; the exact split depends on the model's tokenizer.
How are Open Weights updated?	Open-weight models are updated by releasing a new set of model weights—often after additional pretraining, fine-tuning, or safety training—rather than automatically changing the weights on your machine
is a punctuation mark one token? like comma counts as one token?	Often yes, a comma is represented as one token, but the exact tokenization depends on the model's tokenizer and surrounding text.
Average api cost include both input /output tokens ?	Yes, API cost usually includes both input tokens (your prompt and context) and output tokens (the model's response), although some providers price them separately.
word to token conversion should depend on embedding stretagy ?	No, word-to-token conversion depends on the model's tokenizer, not its embedding strategy; embeddings convert tokens into vectors afterward.
should we avoid punctuations in prompts? since it leads to exhaustion?	No—don't avoid punctuation just to save tokens; it usually improves clarity, and the small token cost is rarely significant compared with long prompts or documents.
when I am making a call, my input may not always be 1M per question, but would I be still charged $2 per call (GPT 4.1)	if GPT-4.1 is priced per token, you are charged for the actual input and output tokens used, not a fixed $2 per call; check the provider's current pricing and any minimum or platform fees.
These Input Output token rates are for explanation or the carges are applied like that only. I have always used free so not aware of this?	They are actual usage charges, not just explanations: the provider multiplies your input and output token counts by their respective per-token rates, though free tiers, credits, minimums, or platform fees may also apply.
Considering output cost is much higher, so is it always better to specify the output size of precise context?	Yes—set an appropriate maximum output-token limit and request concise, structured responses, but allow enough space to complete the task accurately; output pricing is usually higher per token, not necessarily the largest total cost.
Is there any documentation or book for beginners? I'm new to many of these concepts and terminologies.	Yes—good beginner resources include: "AI Engineering" by Chip Huyen — practical LLM concepts, RAG, agents, evaluation, and deployment. "Hands-On Large Language Models" by Jay Alammar and Maartje Grootendorst — beginner-friend
We set max token values while calling LLM APi's , does the token rate gets divided 50% input and 50% output	No—the max output-token value is only a limit, not a 50/50 allocation; you are charged separately for the actual input and output tokens generated, according to the provider's rates.
Practically how to we implement batch mode?	In practice, batch mode groups many independent API requests into a single batch job that is submitted asynchronously and processed together for higher throughput and lower cost.
can we use batch mode to run the scheduler jobs and fetch some data over internet and get some data , is there any time limit like batch would give data in 1 hour or there is no hard limit ??	Yes, scheduled jobs can collect data and submit batch LLM requests, but the time limit depends on the provider's Batch API—some offer an "up to 24 hours" completion window, so check the specific documentation; it is asynchronous, not guaranteed to finish in exactly one hour.
Apologies for posting my question in wrong section earlier. Will we continue to use outdated models throughout these classes? My question is purely from educational standpoint and NOT usage. I understand using older models can be benefitial too but what about studying/education? Is there any difference in studying newer models vs old models?	From an educational perspective, older models are still valuable because they teach the same core LLM concepts and workflows that apply to newer models, with only the capabilities and performance improving over time.
ok, suppose we use an lllm for intent classification, the output by it would be one word(a category). what will that cost economics of a frontier model(for just intent classification) vs an older model(for an entire output) and would models are better for intent classification, does newer models are usually better?	For intent classification, a small or older model is often far more economical and accurate enough, because the output is short and the task is narrow; newer frontier models are usually more capable but may not justify their higher cost.
Are video lessons and blogs enough to steadily progress with weekly learning materials or do we need books as imperative to compete effectively since this programme is like a WILP?	Video lessons and blogs can support steady progress, but for a WILP-style program, use the weekly materials as the core, add one structured beginner-friendly book or course, and reinforce learning through hands-on projects and regular revision
Can we set maximum response token size per request?	Yes, most LLM APIs let you set a maximum output-token limit per request, such as max_tokens or max_output_tokens, to control response length and cost.
Are only the system prompts cached? What about when i ask very common questions like "How to print on console in Java?" Do such prompts also fall under prompt caching too?	Prompt caching usually applies to repeated exact or near-exact prompt prefixes, often including system instructions and shared context—not because a question is common; a one-off Java question normally is not cached unless the provider explicitly supports semantic caching.

what is batch API? and how this can reduce the cost? could you please expalin prompt caching again?	A Batch API lets you submit many independent requests together for asynchronous processing—for example, classifying 100,000 customer messages overnight. It can reduce cost because providers often offer discounted batch pricing, and it improves efficiency by allowing the provider to group workloads; the trade-off is that results are not immediate.
What are the usecases of Batch API? Who uses it?	A Batch API lets you submit many independent requests together for asynchronous processing—for example, classifying 100,000 customer messages overnight. It can reduce cost because providers often offer discounted batch pricing, and it improves efficiency by allowing the provider to group workloads; the trade-off is that results are not immediate.
Is TCO talking about the price break up of how budget gets SPLIT in AI based system	TCO (Total Cost of Ownership) estimates the complete cost of an AI system over time—not just the API price—including infrastructure, model usage, data, development, monitoring, security, maintenance, and support.
If I want to LLM to cache my project architecture and project techstacks then how can I do it?	Use a versioned project-context document plus RAG, and optionally provider prompt caching for repeated context; don't rely on the model permanently "remembering" it unless you fine-tune or use an approved memory system.
how to promt to maximize caching?	To maximize prompt caching, keep a large, reusable prefix identical across requests—system instructions, tools, and stable project context—and put variable user details and questions at the end; use the provider's explicit cache controls if available.
What if I just ask LLM to give me in format of JSON and specifiy the fields before hand. Wont this have lesser output tokens consumed?	Yes, requesting concise JSON with predefined fields usually reduces output tokens by removing explanations, but the JSON keys and syntax still consume tokens, so use short field names and set an appropriate output limit
Are there any special mechanisms for prompt caching ?	Yes—prompt caching typically requires a stable, repeated prompt prefix, such as system instructions or project context; place changing user content at the end and follow the provider's caching rules for minimum size, duration, and eligible models.
i think we had a lecture before this class where the presenter said that the most cost in llm are usually for tokenization and context which runs through every line of input every time, or whenever the context increases, the llm does not have a memory, it just goes through every single word everytime. SO, vene after developemnt of so many good models, we still have the same problem, why don't companies try to solve this problem? won't they make more money if they solve this problem?	Companies are actively solving it, but reducing context processing is difficult because models need relevant context for accuracy, while caching and compression have trade-offs in freshness, memory, privacy, infrastructure complexity, and quality; providers often share efficiency savings through lower prices rather than charging the full amount.
Can you please explain again what prompt caching is? THank you.	Prompt caching means storing and reusing the computation for a repeated part of your prompt—such as system instructions or project documentation—so the model does not fully process that same content from scratch on every request.
Please give live example of how can I force LLM to cache a prompt or part of conext.	You generally cannot force caching unless the provider supports it; use its explicit cache-control feature, keep the reusable prefix identical and at the beginning, place changing content afterward, and verify cache hits through usage or response metadata.
are coding models also probalistic?	Yes, coding models are probabilistic: they generate code by predicting likely next tokens, so the same prompt can produce different outputs unless settings such as temperature are reduced or randomness is disabled.
Hallucination rate more--> does it mean that wastage of cost with no expected or correct responses? is it disadvantage of LLM?	Yes, a higher hallucination rate means more responses contain unsupported or incorrect information, which can waste token costs and require rework; it is a key limitation of LLMs, especially when accuracy is critical.
WHats the difference between prompt caching and grounding data? Is it the same?	Prompt caching reuses repeated prompt content to reduce processing cost and latency, while grounding supplies trusted external information—such as documents or database results—to improve factual accuracy; they are different and can be used together.
for LLM how hallucination can be reduced is it depend upon vendor who is training the model?	Hallucination reduction depends on the model provider's training and alignment, but also on your implementation—use RAG with trusted sources, precise prompts, structured outputs, tool verification, low temperature, citations, and human review.
if we use low temperature parameter then response will be more determinisitcs then in that case do we require LLM to be used?	Yes—low temperature makes an LLM's output more consistent, but it remains useful because it can still understand language, reason, summarize, classify, and generate content rather than merely following fixed rules.
Should we as a practive always give date, should LLM not automatically check if the information requested is date based?	Yes—include an explicit date or "as of" time for time-sensitive requests, because an LLM may not automatically know that freshness matters or have access to current information.

for hallucinations, can we use differnt models(agents for same input context, the agents question each other and can verify the info being outputed), but wont that cause problem as differnt models ahve differnt tokenization method/structure? or is there a way to normalize tokens(as in the ned there are just numbers)?	Yes, multiple models can cross-check one another, but have them exchange plain text or structured JSON, not tokens; tokenization differences do not matter because each model tokenizes the shared output independently, though correlated training data means agreement is not proof of correctness—use trusted sources or deterministic tools for verification.
currently there is lot false news, false explanations , theories etc will this impact general models , how is the model training data validated , as its vast.	Yes, misinformation in public data can affect LLMs; providers reduce it through source filtering, deduplication, quality scoring, human review, trusted references, and post-training, but validation is imperfect and models can still learn or reproduce false claims.
Can LLM search on internet on his own , if we add a propmt and if not does it store all the trained data and training inputs if not connected to internet ??	An LLM cannot search the internet unless it is connected to a browsing/search tool; without internet, it uses only its learned weights and current prompt, and it does not automatically store your inputs or update its training data unless the application explicitly logs or trains on them.
So, while traning an LLM, we use lossy compression for billions of data... What about RAG, does it do a lossy or lossless, and that is why we are seeing LLM hallucinating after using RAG as well	Training is not simply “lossy compression”; it learns statistical patterns in weights, while RAG retrieves chunks that may be incomplete, poorly ranked, or ambiguously interpreted, so hallucinations can still occur even when the source documents are accurate.
Gurav, your quick replies look like AI-generated answers that miss the point. Please request Prof. Rajat to show us a live, practical example of prompt caching if possible	please post in meeting chat for request live.
if references are available with both correct and false information, how LLM will consider which is more reliable? Any mechanism?	LLMs use learned patterns and retrieval/source-ranking signals, but they cannot reliably determine truth automatically; use authoritative sources, metadata and recency ranking, citations, cross-checking, and human or tool-based verification.
Does LLM search tool on web uses the SEO rankings to get the data and similar to web search or have different mechanisam to gather data and different algorithm and what should websites should focus to come in the LLM search rankings	LLM web-search tools usually combine search-engine indexes, ranking algorithms, retrieval, and sometimes direct page extraction—not necessarily SEO ranking alone; websites should prioritize accurate, authoritative, crawlable, well-structured, up-to-date content with clear citations and strong technical SEO.
if llms are not trained properly and no RAG is there chances of more hallucination?	Yes—an inadequately trained model without RAG generally has a higher risk of hallucination because it lacks reliable knowledge and cannot consult trusted, up-to-date sources.
is it possible to use a multiple models to decreace the risk of hallucination? Just like how we have multiple humans for one single job so that other person can verify the work..	Yes, using multiple LLMs in an ensemble—where one or more models independently verify or critique another model's output—can significantly reduce the risk of hallucinations, much like multiple humans reviewing the same work.
How do we decide when we go greedy o the algorithm or not , does it have to do with what layer of weights we are trying to fine tune for a given application ?	Whether decoding is greedy or not is decided at inference time based on the application's need for accuracy versus diversity, and it is independent of which model weights or layers were fine-tuned.
can you please repeat the 3 types of hallucination	Factual hallucination: Invents or states false facts. Faithfulness hallucination: Misrepresents or contradicts the provided context. Instruction hallucination: Fails to follow the user's instructions or constraints.
Who gives this ECE ratings? And how can it be trusted?	ECE is usually calculated by researchers or benchmark providers from a model’s confidence scores and labeled test data, and it is trustworthy only when the dataset, scoring method, model version, prompts, and evaluation code are transparent and independently reproducible.
Is there a way to integate output calibration report using any tool or SDK while testing prompts?	Yes, frameworks such as OpenAI Evals, LangSmith, TruLens, and DeepEval can be integrated into your prompt-testing pipeline, although true calibration metrics like ECE require tasks with ground-truth labels rather than prompt outputs alone.
which models are them well calibrated? how to check ECE fro model?	Well-calibrated models are those whose confidence scores closely match their actual accuracy, and calibration varies by model, task, and fine-tuning rather than by model family alone. You can measure this using Expected Calibration Error (ECE) by comparing predicted confidence with observed accuracy across confidence bins, where a lower ECE indicates better calibration.
Is there an inbuilt mechanism or tools to check ECE ?	No, most LLM APIs do not provide a built-in ECE calculator; ECE is typically computed offline using evaluation libraries (e.g., scikit-learn-based implementations, TorchMetrics, or custom scripts) on predictions with ground-truth labels.
suppose we have any LLM for example qwen then how to check ECE of it? could you please name the LLMs which has lowest ECE and can be considered as well calibrated ?	There is no reliable universal list of LLMs with the lowest ECE—calibration varies by model version, task, prompting, decoding, and calibration method; measure it under your own conditions and consider temperature scaling or conformal calibration.
yeah, llms are supposed to be good at undersatnding human texts only right? they are not good for statistical analysis of numbers, right? so, i have seen these llm based quant trading projects, then these cannot be trusted right?	LLMs can interpret text and assist with data analysis, but they are not inherently reliable statistical or trading engines; quant-trading systems should use validated quantitative models, market data, backtesting, risk controls, and human oversight rather than trusting LLM predictions alone.

In what scenario, we should use high temperature?	Use a high temperature when you want the AI to give more varied and creative answers—for example, asking it to suggest 10 different names for a new café rather than one predictable name.
NLI Based= Please rlsborate. Also setting temperature in halucianation means?	temperature controls how adventurous the model's wording is; a lower temperature usually makes answers more consistent and may reduce mistakes, while a higher temperature encourages variety but can increase the risk of unsupported answers. Lower temperature cannot guarantee truthfulness.
What do you mean by temperature in case of LLMs?	Temperature is an LLM parameter that controls the randomness of token selection, where lower values produce more deterministic outputs and higher values produce more diverse and creative outputs.
what are reasoing models (if it is based on probablility) what does it mean when llm is thinking?	Reasoning models are LLMs trained to perform multi-step inference, where "thinking" refers to iteratively generating and evaluating intermediate computations to improve the probability of producing a correct final answer, rather than conscious thought.
What is NLI model?	An NLI (Natural Language Inference) model is an AI model that determines whether one statement entails, contradicts, or is neutral with respect to another statement.
But this NLI model need more responses to cross check, means more token and cost. So implementing NLI Model will be expensive but more reliable, right?	Yes, using an NLI model generally increases token usage and cost because it performs additional verification, but it provides more reliable outputs by detecting unsupported or contradictory claims.
in halucination topic- detection strategy-Sample Multiple Outputs with temparature >0 Did not understand hoe to do this Practically. Kindly give eg	Example: Ask the same question (e.g., "Who invented the telephone?") 5 times with temperature = 0.8; if the model gives different answers across runs, the inconsistency indicates the response may be hallucinated and should be verified.
Won't scoring every fact consume more token ? How to optimise in this scenario ?	Yes, scoring every fact increases token usage, so optimize by verifying only high-risk or low-confidence claims, or by batching multiple facts into a single verification prompt.
5 times with temperature = 0.8; with temparature = 0.8 means? we write like this in the promt?	No—`temperature=0.8` is an API/model parameter (not part of the prompt) that you set in your code or model configuration, and then send the same prompt multiple times to compare the outputs.
if we ask to get output in JSON? does JSON needs more tokens in generation then it costs more right?	Yes, generating JSON typically uses slightly more output tokens than plain text because of the added structural characters (braces, quotes, keys), resulting in a small increase in cost.
How do we run NLI faithfullness?	To run NLI-based faithfulness checking, split the generated answer into factual claims and use an NLI model to classify whether each claim is entailed, contradicted, or unsupported by the retrieved context.
Mitigation playbook are eventually the default prompts we set when we do api call to LLM probably like a gateway service that tag all these prompts. THey can eventually increase token cost. are there caching techiniquest that help us to balance the token cost and better results from model	Yes, techniques like prompt caching, semantic caching, and response caching can reduce repeated token costs by reusing previously processed prompts or results while still applying mitigation prompts for better reliability.
how do we "split the generated answer into factual claims and use an NLI model to classify whether each claim is entailed, contradicted, or unsupported by the retrieved context."?	We split the answer into individual claims (using an LLM or rule-based method), pair each claim with the retrieved context, and pass these pairs to an NLI model that labels each as entailed, contradicted, or unsupported.
In our first upgrad lecture, we were told that JSON schema increases the number of tokens consumed. So, is it recommended to use the JSON schema in prompts	Yes, JSON schema can increase token usage, but it is recommended when structured, reliable, and machine-readable outputs are important because the small cost increase is often worth the consistency gained.
Mitigation playbook are eventually the default prompts we set when we do api call to LLM probably like a gateway service that tag all these prompts. THey can eventually increase token cost. are there caching techiniquest that help us to balance the token cost and better results from model	Yes—mitigation prompts can add input tokens, but you can balance quality and cost by using prompt caching, short reusable instructions, RAG, smaller models for simple tasks, and routing only difficult cases to larger models
Are there any good practices for constraining output to a json structure, like a pydantic model?	Yes, good practices include using structured output features or JSON schema validation (such as Pydantic models), clearly defining required fields, enforcing types, and validating/retrying invalid responses before use.
In Prevent halucination -Constrain model to structured output(json schema) - kindly elaborate with an example	Example: Instead of asking "Summarize this customer complaint," require the LLM to return a JSON schema like `{"issue": "string", "severity": "low|medium|high", "evidence": "string"}`, which limits the model to provide only defined fields and reduces unsupported responses.
What does "tab completion" mean?	Tab completion is a feature where an AI system predicts and automatically suggests the next words, phrases, or code lines while a user is typing, similar to autocomplete.
Kind request: Please use updated facts in PPTs 🙏	Feedback will be forwarded.
So cursor has a contract with the models and we have to pay per requests to them? So no token calculation etc.?	Cursor typically charges users through its own subscription plans, while the underlying AI model usage may involve provider costs; token usage still matters because model APIs generally price based on input/output tokens, even if users do not see the calculation directly.

LLMs are only trained on open source, there exists a whale of information, standards, security practices etc, in closed source. When vibe coding applications, how can we ensure closed source features like production grade : 1. security 2. standards 3. business knowledge. My question is, how can we bring these closed source practices to a vibe coded app	Use AI-assisted development with curated internal knowledge bases, secure coding frameworks, architecture standards, expert reviews, automated testing, and governance processes to inject enterprise-grade security, standards, and business expertise into vibe-coded applications.
why not codex is not compared here? any reason?	Codex can be part of the solution, but the comparison was about the broader process of importing closed-source practices into development rather than any single coding assistant or model.
What is IP indemnity?	IP indemnity means a company promises to protect you if its product causes a legal dispute because it copied or improperly used someone else's intellectual property, such as code, text, or inventions.
which coding tool will use llm internal only	Cursor, windsurf, etc.
Claud Code is not free to use right?	Yes, Anthropic's [Claude Code] is a paid tool that requires access through eligible Claude plans rather than being a completely free standalone coding tool.
How does tokens to credit conversion happens in windsurf? like how many token for 1 credit? or are there any other params included?	Windsurf credits are not a fixed "tokens per credit" conversion; usage depends on the model, input/output tokens, request complexity, and any cached context, so check Windsurf's current plan documentation or in-app usage panel for the exact calculation.
will the claude -p command create a claude session? will i be able to see it later using the history command?	No, `claude -p` runs Claude in non-interactive print mode and does not create a persistent interactive session that you can later view with the `history` command.
Does "@CodeBase" take into account all the repos in a cursor workspace or single repo?	In Cursor, `@CodeBase` searches the indexed codebase for the current workspace, which can include multiple repositories if they are all opened within the same workspace.
If I ask an AI to analyze a specific file or search the @codebase, does that burn through more input tokens? What's a better, more cost-effective way to do it?	Yes, asking AI to analyze a specific file or search `@Codebase` can consume more input tokens because it may include larger context, so the cost-effective approach is to provide only the relevant files, snippets, or targeted queries and let the AI retrieve the minimum necessary context.
Whic tool or IDE we will going to use in this program ?	The tool or IDE depends on the program setup, but common AI-assisted development choices include OpenAI Codex, Cursor, Windsurf, or Claude Code depending on the workflow and requirements.
vibe code also provide test and devlopment option	Yes it can. but developer needs to verify the same
What does "writing tests" mean? I usually use cursor's plan mode.	tests for the code written by LLM's
out of these two promts shown in slide, which uses use more tokens?	good promt means less tokens.
say i am building independent POC where i have good idea on how user flow and architecture should look like, is it safe to provide that document to have the tool start coding.. because it will generate 100's of lines in one go. If not what would be safer way to proceed	make small task, verify, write tests around that, increase the scope gradually.
In the context of vibe VS disciplined workflow, developers are now using one AI agent to review another AI's code. Is this a viable and sustainable practice?	its oaky to use that but with due diligence
Can we have the live practical example of it?	Yes, there is a hands-on session.
In this slide, when you mean by reject, does it mean I have to re - generate the current prompt with the added constraint or tell the constraint as the next question? why: context exhaustion comes into play here?	Reject means you should regenerate the current prompt with the added constraint (rather than waiting to state it in the next question), because otherwise context exhaustion can cause the model to lose or underweight that constraint in later turns.
Can we use locally installed LLM models as AI pair programmer does those tools work like cursor or github ??	Yes, locally installed LLMs can serve as AI pair programmers through tools like Continue, Aider, or Cline, offering functionality similar to Cursor or GitHub Copilot, though the quality depends on the model and your hardware.
what is prisma means here in thsi slide	In this slide, Prisma most likely refers to the Prisma ORM, a tool that simplifies database access and management in applications by letting developers work with databases using type-safe code instead of writing raw SQL.
Is there any prep work or prerequisite required for us to follow along with the faculty during the practical session?	It will be done in Google Colab.
can you please share github link?	live answered
ERROR: Could not find a version that satisfies the requirement sentencepeice (from versions: none) ERROR: No matching distribution found for sentencepeice	The package name is misspelled: use sentencepiece, not sentencepeice.
how do we check that these are installed?	Professor has already shown in the class. it shows over there only
Is the installation of tiktoken is manadatory in everry project or it will be oonly for interrnal use to know the numberr of tokes used?	No, installing tiktoken in every Colab project is not mandatory; install it only when your project needs to count or split text into tokens, and Colab sessions may require reinstalling it after restarting.
can the text be any text ?	Yes for learning it can be anything.

Whats cl100k_base?	cl100k_base is a tokenizer vocabulary/encoding name. in Python, you load it using tiktoken.get_encoding("cl100k_base").
can tiktokens predict number of output tokens as well or do we use transformers to predict out put token count	tiktoken can count the tokens in a text you already have, but it cannot know the exact output length before generation; you can set a maximum output limit, while the actual length depends on the model's response.
For me it is only 27 tokens.	if same text, model, tokenizers - should be same. the token count for the same text can vary if you use different models or tokenizers; with the same tokenizer, the count remains the same.
Why is token count different for me ? I wrote same text	you changed the tokenizer, text, formatting, or included hidden characters.
mine are 28	if same text, model, tokenizers - should be same. the token count for the same text can vary if you use different models or tokenizers; with the same tokenizer, the count remains the same.
getting different number than what is shown here	you changed the tokenizer, text, formatting, or included hidden characters.
These numbers that are showing up. Are they the number representation of the respective words?	Yes, those numbers are IDs assigned to tokens; each ID represents a word, part of a word, punctuation mark, or space—not necessarily a complete word.
what are different inputs available for get_encoding	cl100k_base # GPT-4, GPT-3.5-turbo, "o200k_base" # newer OpenAI models etc
I got 28 tokens. Is that correct?	differences usually come from hidden spaces/newlines, different tokenizer versions, extra chat-message formatting, or counting the prompt differently from the displayed text. You can open two colab and then try in two different windows to verify
I got 24 as length. However how can i see how words are broken into	use decode_single_token_bytes
I wrote the same text and generated the tokens, how come my tokens are only 23 in number. Can there by such high variation between each executions?	you might changed the tokenizer, text, formatting, or included hidden characters.
text = """ Artificial Intelligence (AI) is transforming software engineering by enabling AI-native development, automated code generation, and intelligent software testing. """	please post in zoom chat to get the exact text. however idea is to make you hands on not to focus the exact text.
Demo: For me number of token for same text is 28. Please explain why?	you changed the tokenizer, text, formatting, or included hidden characters.
I got 28 ntot 43	if same text, model, tokenizers - should be same. the token count for the same text can vary if you use different models or tokenizers; with the same tokenizer, the count remains the same.
is cl100k following the same rule 1000 words 1333 tokens approx?	cl100k_base does not follow a fixed rule like "1,000 words equals 1,333 tokens"—token count depends on the language, word length, punctuation, spaces, and formatting.
it wopuld be good if you can paste the exact text here so that we can copy	live answered
Demo: We used exactly same code abd text then why it is different token number?	differences usually come from hidden spaces/newlines, different tokenizer versions, extra chat-message formatting, or counting the prompt differently from the displayed text. You can open two colab and then try in two different windows to verify
Wil the tab spaces are also considered as a token?	Yes, tabs and spaces can count as tokens, either alone or combined with nearby text, depending on the tokenizer.
.decode is not working 'method' object is not subscriptable	may be syntax issue. pls check your code
why is million written like 1_000_000 ? Is it the syntax ?	Yes
why diving input token /100000	price is quoted per 100,000 tokens
How to type rupee symbol?	Windows: Press Ctrl + Alt + 4 or use ₹ via copy-paste. Mac: Press Option + 4 in some keyboard layouts.
As each model has the algorithm to generate tokens and can vary from provider to provider how does token calculating libraries work and trustable to calculate the real cost ??	Token libraries use a model's published tokenizer to count tokens before sending a request, but exact billing may also include hidden chat formatting, tools, cached tokens, or provider-specific rules—so treat the count as an estimate and confirm the API's usage details.
where is the code being posted?	Will share in session resources on Prism
It means we need to give the model's name in the token calculating libraries to have the estimated cost ??	Yes—provide the model name so the library can use the correct tokenizer and pricing; otherwise, its token and cost estimate may be inaccurate.
cannot see the demo window on zoom	please check your conenction or rejoin. its visible.
not sure you understand Gaurav, Where is the professor pasting the code?	this is online editor. I've asked the team to answer this if you are looking for codes.

why fix size of plot cant it should be dynanic , tokens can be more is there way to do that?	Yes, make the plot size dynamic by calculating its width from the number of tokens: import matplotlib.pyplot as plt n = len(token_ids) width = max(10, n * 0.35) plt.figure(figsize=(width, 5)) plt.plot(token_ids) plt.show()
how to do that?	answered already
can we expect different responses eventhough old and new prompt text is same	Yes
one more taken can be saved if you skip the "." in new prompt	possibly
where is github shared?	https://github.com/Sabyasachi123276/Hands_on_Session_Foundation-Models-Token-Economics-AI-Pair-Programming-Module-1.1-/blob/main/Hands_on_Session_Foundation_Models%2C_Token_Economics_%26_AI_Pair_Programming_(Module_1_1).ipynb